Databases

Introduction

Studying Method

- Self study, focus on the fundamental concepts and refer to personal experiences
- Doing exercises
- Developing a project or attending the exercises during the lectures, using a proper tool (such as *DB2, SQLServer, Oracle, PostgreSQL, MySQL, MS Access, …*)

Data base

- An organized set of data that supports on carrying out specific activities (within an institution, an enterprise, an office, a human)

Point of view

- Methodological
- Technological

Contents

- Models for organizing data
- Languages for using data
- Systems for handling data
- Database design methodologies

Information system

- A component of an institution that handles the interesting pieces of information (e.g. in order to pursue the corporate goals)
- Each institution has its own information system, that could be not explicitly identifiable within its organization
- The information system supports other “sub-systems”. For this very reason it should be studied inside its operative context.

Handling the information

- Gathering, acquisition
- Storing, preservation
- Elaboration, trasformation, production
- Distribution, communication, exchange

Information system and automation

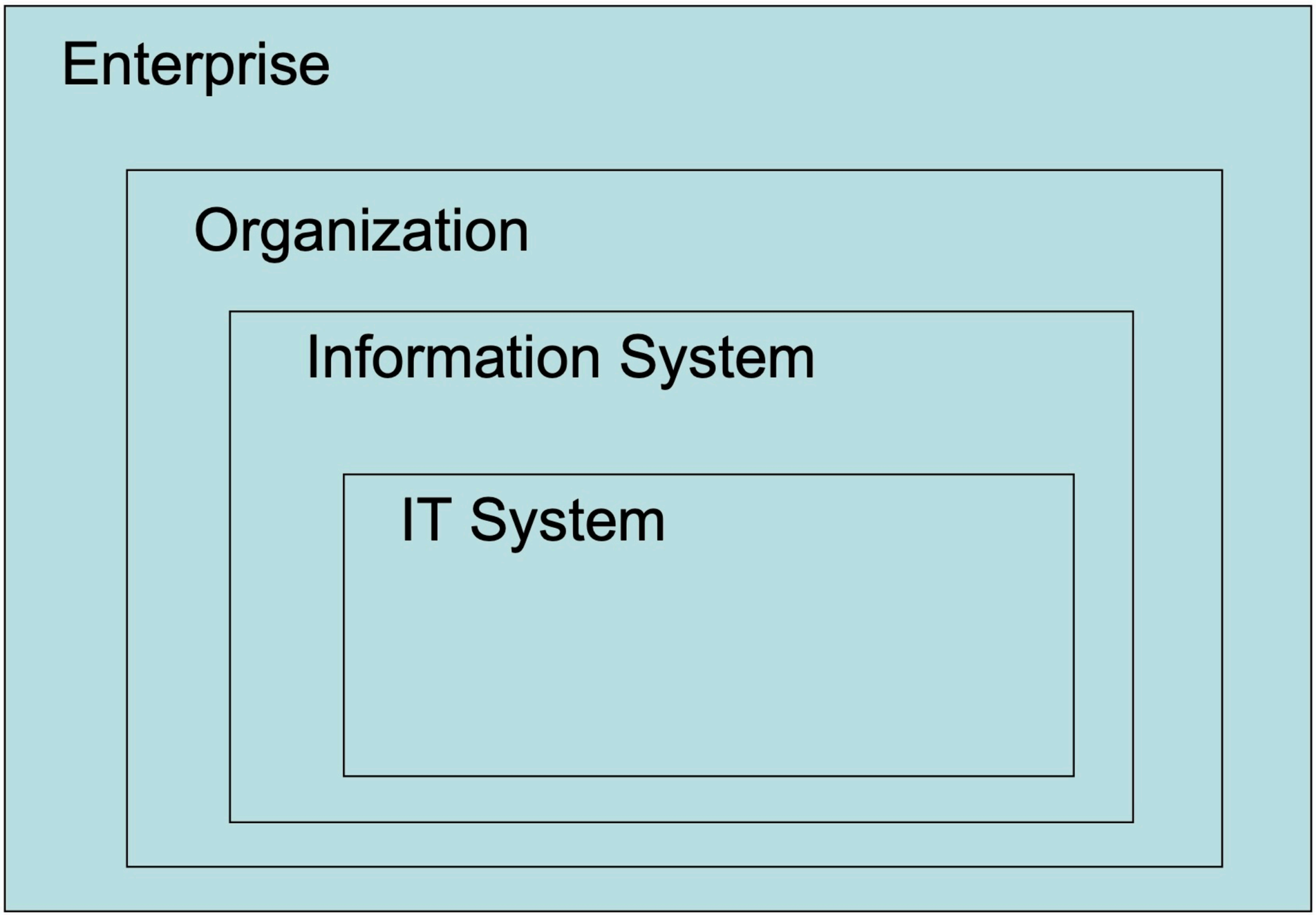
- The idea of “information system” is independent from any computer automation:
  - Some organizations have the only mission of handling data (e.g. demographic recordings and banking services) and they existed for centuries

IT System

- One of the automated parts of the Information System:

  It's the information system component that handle information using computer technologies

IT System



Handling the information

- Different human activities could represent the pieces of information differently:
  - informal ideas
  - natural language (written or word of mouth, formal or conversational, in different languages...)
  - drawings, plottings, diagrams
  - numbers and codes
- Such information has different storage media:
  - Human brain, paper, electronical devices

Information and data (1)

- Within (e.g.) IT Systems, the information is roughly expressed as data

Information and data (2)

(Oxford Dictionary)

information: Facts provided or learned about something or someone.

data: Pieces of information. A fixed starting point of a operation.

Information and data

Mon-Fri Saturday Holiday

- What does those numbers mean?
- Road signs in Finland; they are hours
- But what is the difference between them?
- The datum is useless without its “interpretation”

Handling information

- Data is the result of the process of organizing, coding and storing of information
- E.g., within demographic registries when referring to citizens
  - Wordy descriptions
  - Name and Surname
  - Personal pieces of information
  - Personal Tax Code

Why data?

- It is difficult to precisely represent very rich pieces of information and knowledge
- Data are a strategic resource, because they are more stable than other representation (business processes, technologies, human roles):
  - e.g., data in banks and demographic registries

Databases

(generical meaning)

- An organized set of data that supports on carrying out specific activities of an entity (an institution, an enterprise, an office, a human)

(specific meaning)

- Set of data handled by a DBMS

DataBase Management System (DBMS)

- Any system handling data collections that are:
  - big
  - persistent
  - shared
- While ensuring:
  - privacy
  - reliability
  - efficiency
  - effectiveness

DBMS

- Software as a product and as a service (complessi) available on the market:
  - DB2
  - Oracle
  - SQLServer
  - MySQL
  - PostgreSQL
  - Access
  - BigQuery
  - SQLite

Databases are ... big

- their size are (by far) greater than the computing systems' main memory
- Only the devices' physical limit is considered
- Some example of big sizes
  - 1-5 Terabyte (transactional data)
  - 30-50 Terabyte (decision data)
  - 500-800 Terabyte (scientific data)
  - 100 billions of records

Data bases are ... persistent

- their lifetime is independent from the lifetime of the computer processes using such data

Databases are … shared

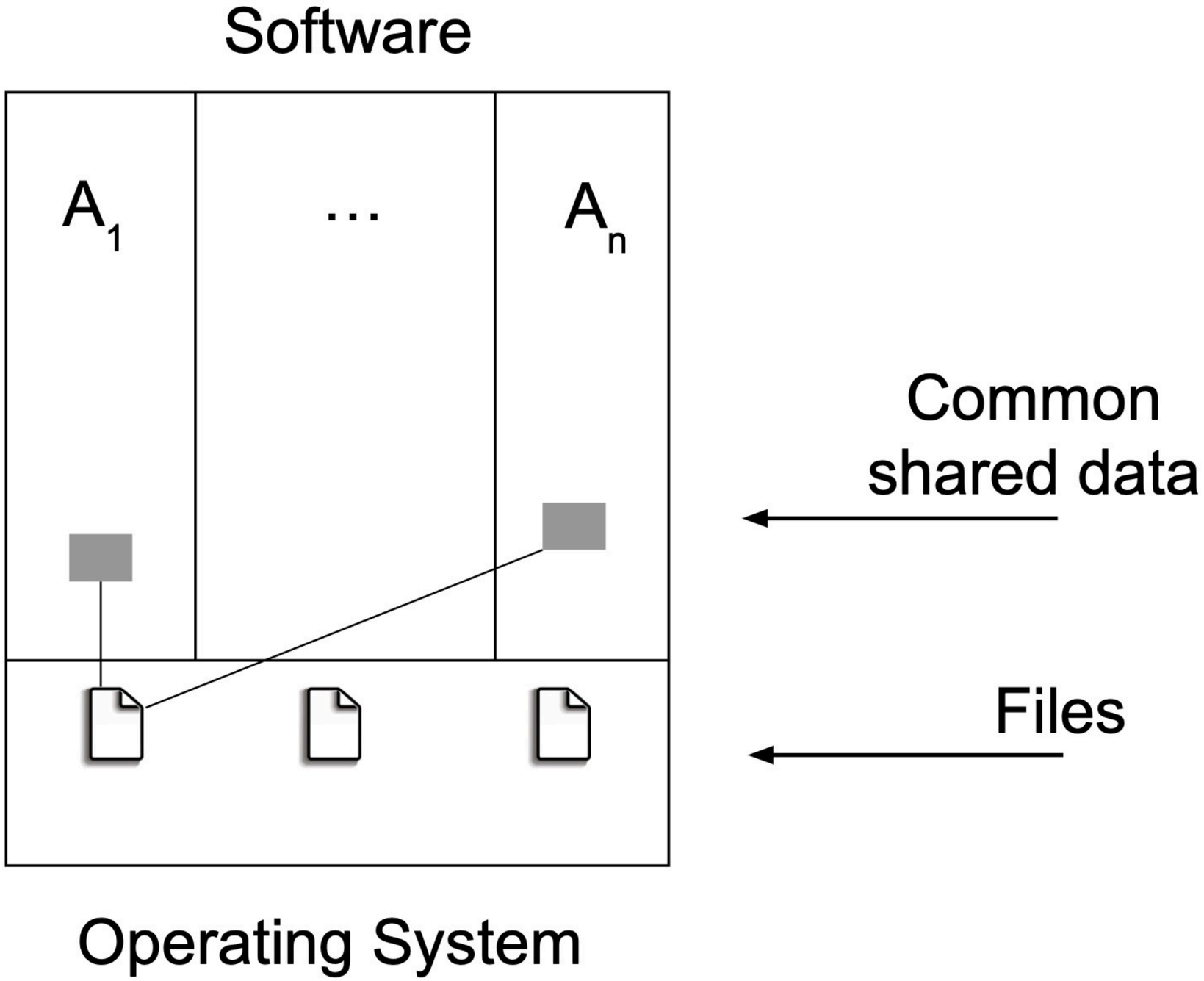
- Every (big) corporation is sliced in different areas, e.g. it could carry out different activities
- Each area/activity has a information (sub)system (that is not necessarily detached from the central one)

Problems

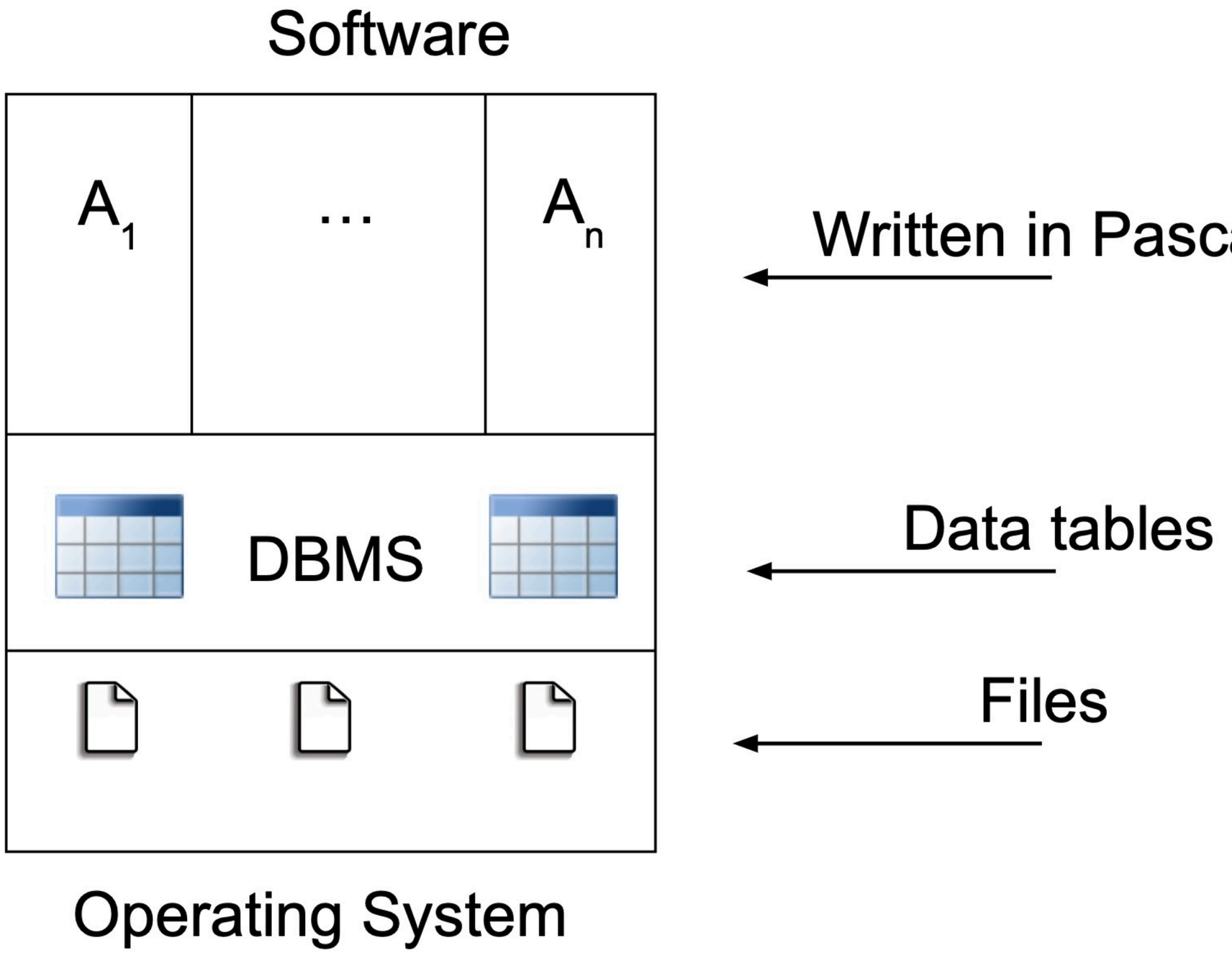
- Redundancy…
  - same data appears many times
- …could cause inconsistency:
  - different descriptions could not match

Let’s see how Data Management evolved through time

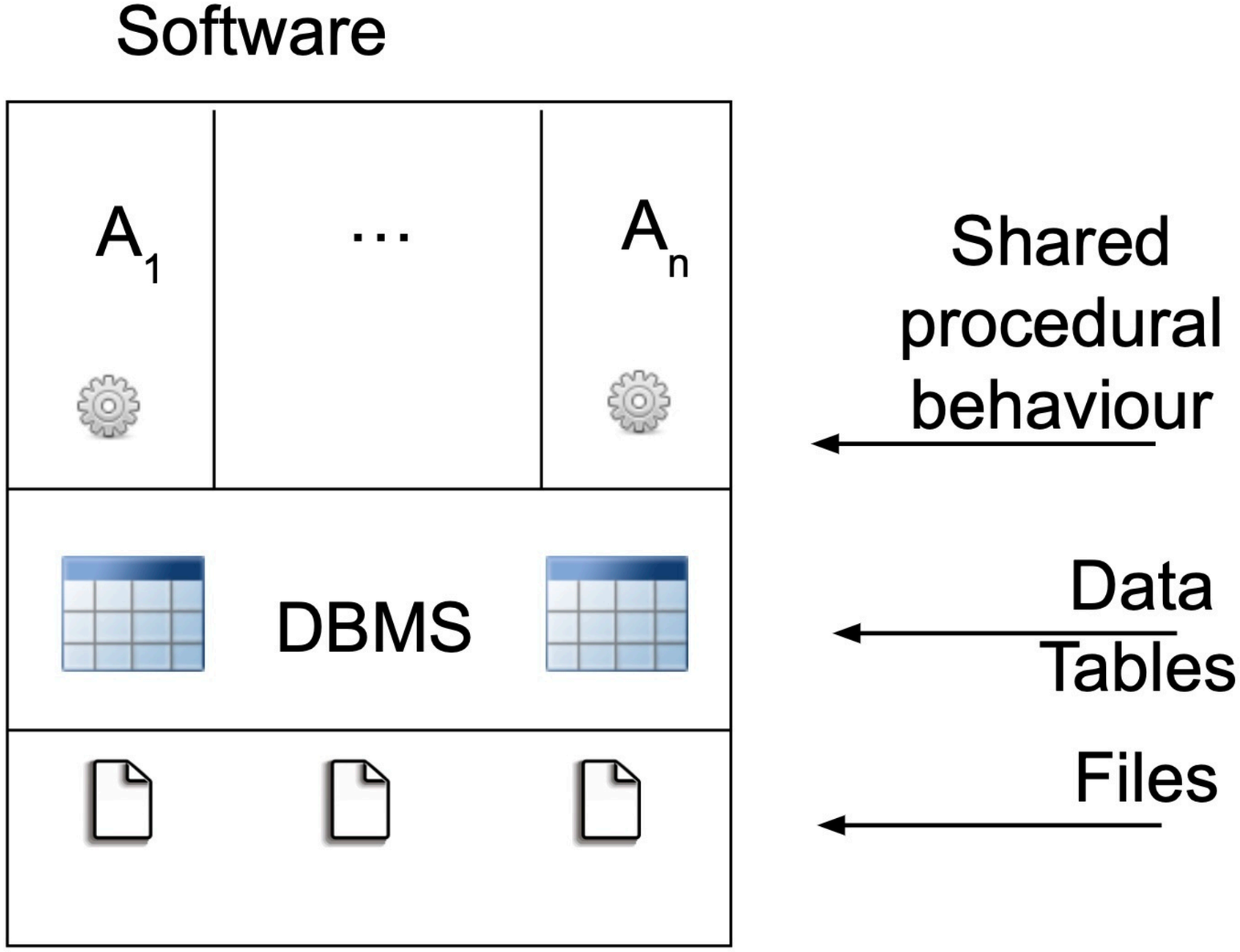
The 70s: no DBMS



The 80s: first DBMSs



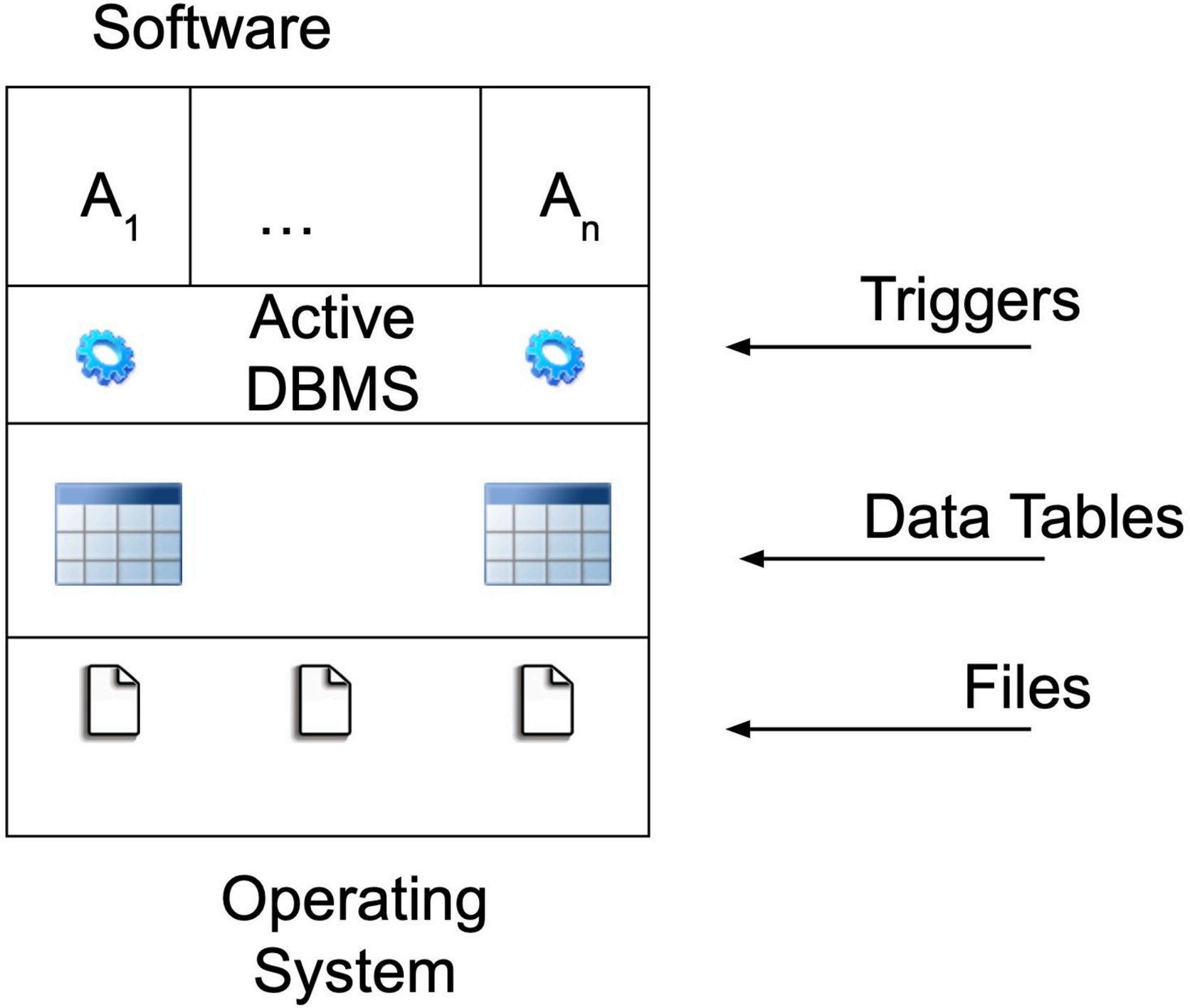
The 90s: the procedural behaviour is shared among the software applications



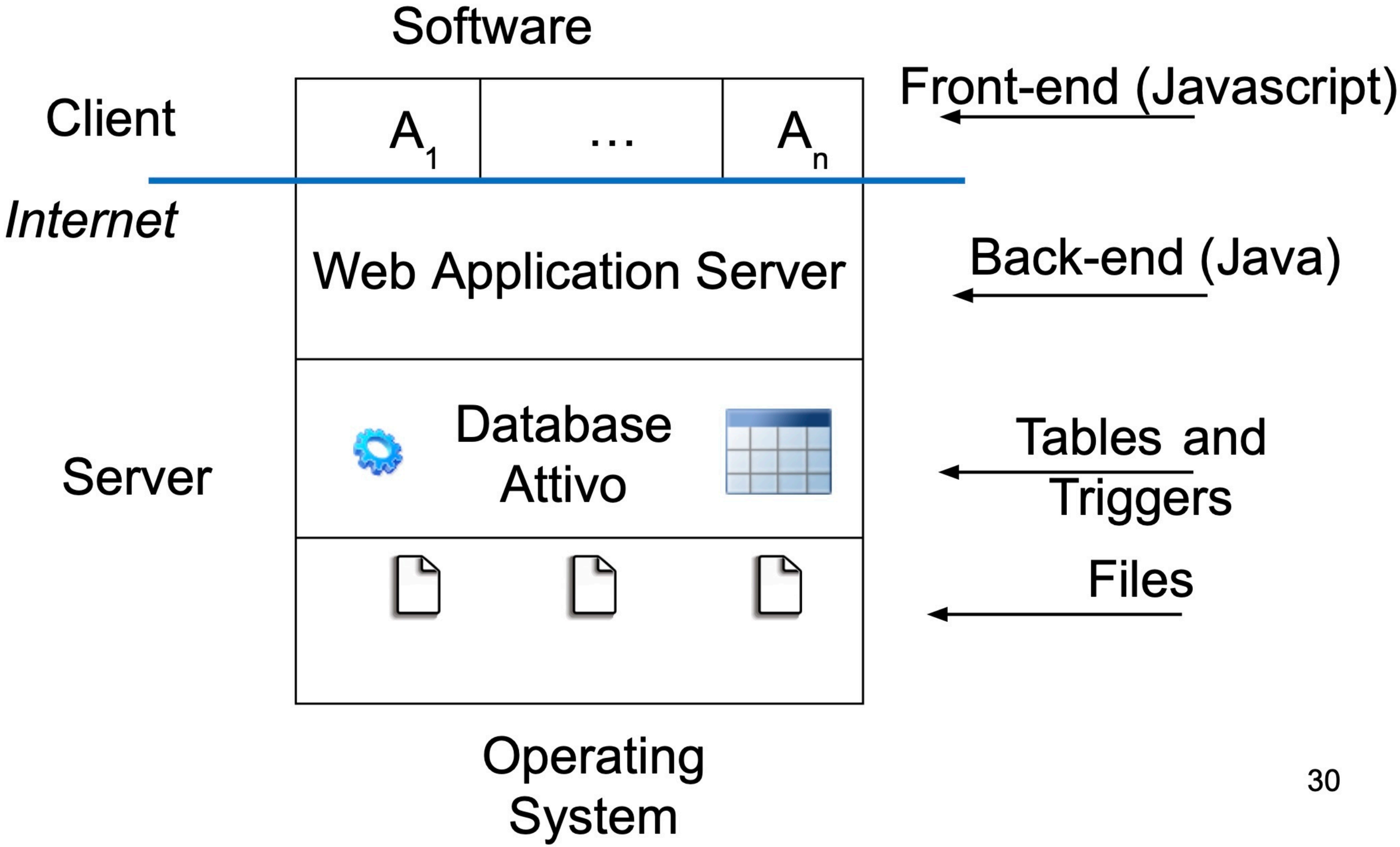
Advances in DBMSs: Stored procedures

- Stored procedures have been introduced to share the common procedural behaviours between different software.
- Stored procedures are not standardized and are affected by the problem of impedance mismatch (we'll see later) with the language used to express such procedures
- As a result specific rules (*triggers*) have been introduced to model the procedural behaviour shared among different software and are handled by the DBMS itself.

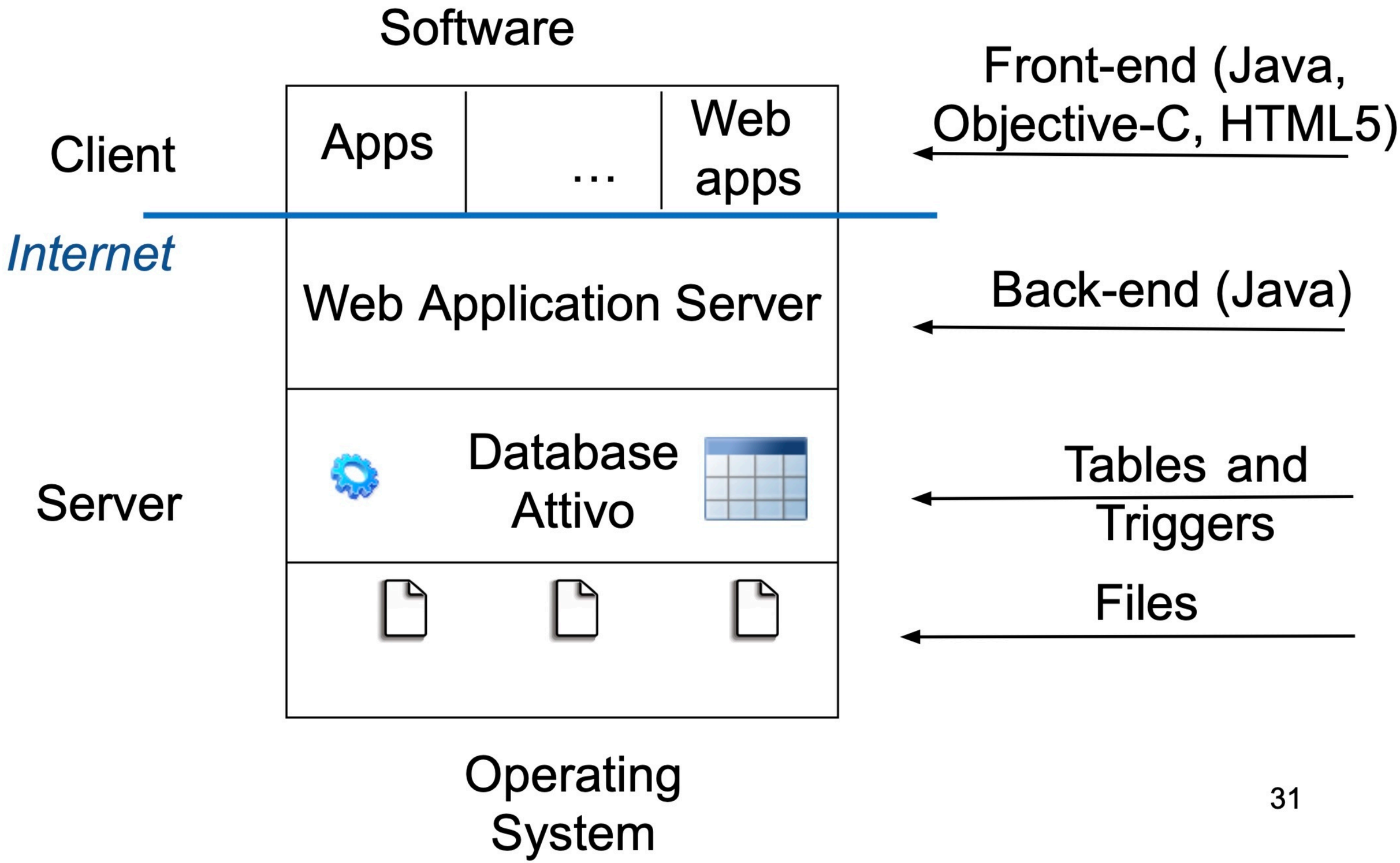
The 90s: Active DBMS



Advance in the 2000s



The 2010s: mobile apps



Databases are … shared

- A data base is an integrated and shared resource among the software applications
- As a consequence
  - Different activities build up on shared data:
    - requires permission mechanisms
  - More users access shared data:
    - requires concurrency checks

Databases preserve ... privacy

- We could implement permission mechanisms such as:
  - "*User A is allowed to read all the data and to edit data X*"
  - "*User B is allowed to read data X and to edit data Y*"

Databases preserve ... reliability

- Reliability (w.r.t. data bases):
  - are tolerant w.r.t both hardware and software failures
- A database is a precious resource and, consequently, should be preserved for a long time to come
- Crucial technique:
  - Transactions handling

Transaction

- A set of non-detachable operations (*atomic*)…
- that are correct even on a concurrent system…
- …with final effects

Transactions are … atomic

- A sequence of related operations:
  - "*Move the money from the bank account A to B: either the withdraw is performed on A and transferred to B or no operation is carried out*"
- … either is performed as a whole, or is not performed at all:
  - "*either the withdraw is performed on A and transferred to B or no operation is carried out*"

Transactions are … concurrent

- Concurrent transactions must be coherent.
  - *If two different checks are issued on a same bank account and are cashed contemporarily*

    … you must not miss one of them
  - *If two different agencies want to book the same (available) sit on a train*

    … you must not book it twice

The effect of transactions are permanent

- The commit of a transaction requires that the final result should be logged and tracked, even in a concurrent context or when failures occur.

Databases must be … efficient

- They try to use profitably the memory (*primary* and *secondary*) and time (of *running* and *response*) resources.
- Since DBMS provide many operations, they could be implemented inefficiently. For this reason there are huge investments and competition.
- Efficiency is also a result of software quality

Databases must be … effective

- Databases try to improve the activities of their users, by providing powerful and flexible functionalities:
  - a frequent topic of the lectures is how DBMSs pursue effectiveness

DBMS vs file system

- Big collections of data could be also handled by simpler systems, such as standard Operating Systems' file system
- File systems provide a coarse grained access to data: "all or nothing"
- DBMSs extend the file system's features, by providing more services and in a more integrated fashion

Describing data

- Each software (that does not rely on a DBMS) contains an internal description of the file structure that is going to be processed. As a drawback, multiple software could have multiple data description: this situation generates inconsistency within data representations.
- In DBMSs, a portion of the database (the catalogue or dictionary) contains a centralized description of the data: the same description is shared among all the software applications.

Describing data in DBMS

- Data are described at different levels of abstraction
  - Data are not dependent from their physical representation:
    - Programs use a high level representation, independent from the data representations at a lower level. As a consequence, a change in the lower level does not require changes in the programs.
  - This is more precisely obtained through the data model concept

Data Model

- A set of constructs that are used to organize and describe the behaviour of the data
- Crucial component: providing structure to data (via type constructor)
- The data model provides some default data type constructors, as it usually happens on current programming languages.
- E.g.: the relational model provides the constructor relation, allowing to define a set of homogeneous records

Organizing data in a database

Scheduling

Lectures	Lecturer	Room	Time
Calculus I	Luigi Neri	N1	8:00
Databases	Pier Rossi	N2	9:45
Chemistry	Nicola Mori	N1	9:45
Physics I	Mario Bruni	N1	11:45
Physics II	Mario Bruni	N3	9:45
IT Systems	Pier Rossi	N3	8:00

Organizing data in DBMS

A schema

Scheduling

Lectures	Lecturer	Room	Time
Calculus I	Luigi Neri	N1	8:00
Databases	Pier Rossi	N2	9:45
Chemistry	Nicola Mori	N1	9:45
Physics I	Mario Bruni	N1	11:45
Physics II	Mario Bruni	N3	9:45
IT Systems	Pier Rossi	N3	8:00

Some istances

Schema and instances

- In every database there exist:
  - the schema: it is time invariant and describes the data's structure (intensional aspect)
    - e.g.: tables' headers
  - the instance: the actual values that could rapidly change (extensional aspect)
    - e.g.: the tables' body

Two different aspects

- **Logical** Models
- **Conceptual** Modelling

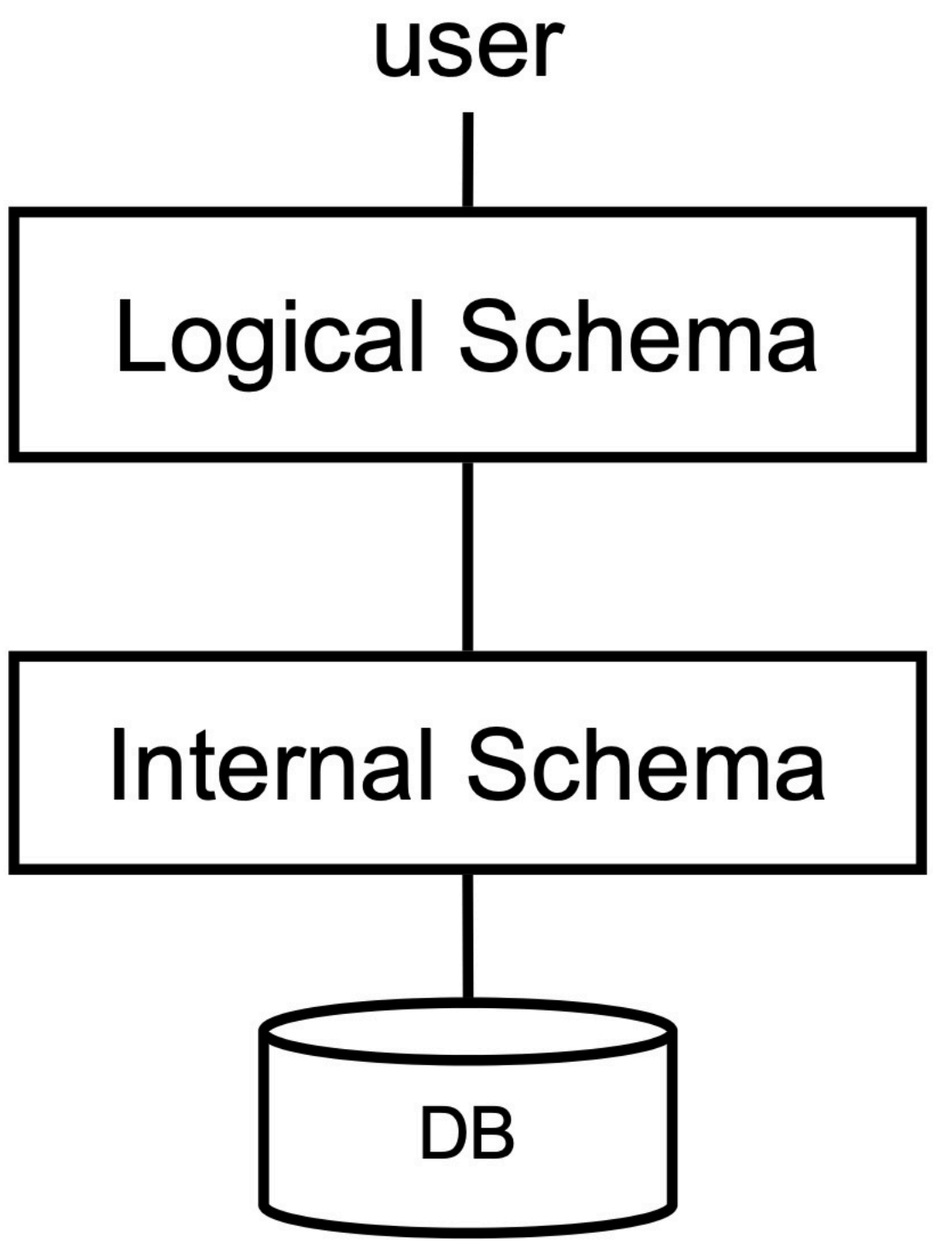
Logical Models

- DBMS use such models to store and organize the data
  - used by software at a higher level
  - independent from the physical representation
- e.g.: relational, lattice, hierarchical, object, XML

Conceptual Modelling

- They represent data independently from each specific system
  - they describe concepts from the real world
  - they are used in the preparatory phase of a project
- The most diffused data model language is Entity-Relationship

Simplified Architecture of a DBMS



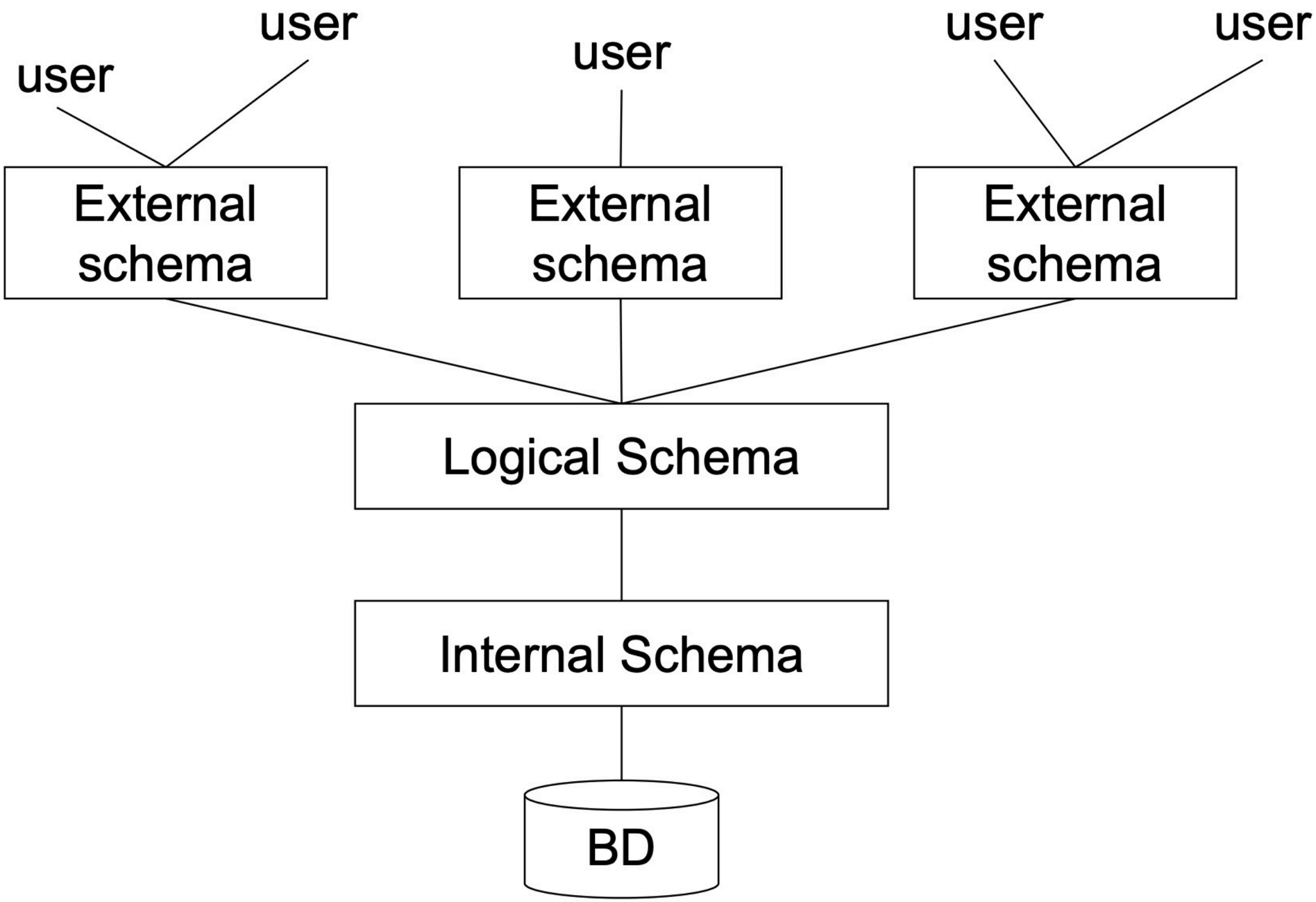
Simplified Architecture of a DBMS: schema

- Logical schema: describes the structure of the database through the logical data model (e.g., the table's structure)
- Internal (or physical) schema: it represents the logical schema at the storage level using specific *raw* data structures (such as files, records and data pointers)

Data independence

- The data representation at the logical level is independent from the one at the physical level:
  - a table is handled “as it is” independently from its underlying representation (that could even vary over time)
- For this very reason in this course we will see only the logical level and not the internal one.

Standard three-tiered ANSI/SPARC architecture



ANSI/SPARC architecture: schema

- a logical schema describes the structure of the database through the “main” logical data model
- an internal schema represents the conceptual level at the storage level using specific *raw* data structures (such as files, records and data pointers)
- an external schema describes part of the database in a logical model (partial “views”, derived tables, even from different models)

A View

Lectures

Lecture	Lecturer	Room
Databases	Rossi	DS3
Systems	Neri	N3
Networks	Bruni	N3
Controls	Bruni	G

Rooms

Name	Building	Floor
DS1	OMI	Ground
N3	OMI	Ground
G	Pincherle	First

LectureRooms

Lecture	Room	Building	Floor
Systems	N3	OMI	Ground
Networks	N3	OMI	Ground
Controls	G	Pincherle	First

Data independence

- Is the consequence of having different layers
- The data access is provided by the external level (that could coincide with the logical one)
- two types of independence:
  - Physical data independence
  - Logical data independence

Physical data independence

- the logical and external levels have a representation which is independent from the physical level
  - a relation is “handled” in the same way independently from its physical implementation
  - The physical implementation could undergo some changes without the need of changing the software

Logical data independence

- The external level is independent from the logical level
- Edits and Updates on “views” do not require any change at the logical level
- “Transparent” edits of the logical schema keep the external schema unaltered

Database languages

- Another aspect of the database effectiveness is given by its availability through different languages and interfaces
  - "interactive" textual languages (**SQL**)
  - statements (SQL) injected in a host language (Pascal, Java, C ...)
  - statements (SQL) injected in an *ad hoc* language with other features (e.g. plottings, printing tables)
  - user friendly interfaces (with no textual interface)

SQL, an interactive language

Lectures

Lecture	Lecturer	Room
Databases	Rossi	DS3
Systems	Neri	N3
Networks	Bruni	N3
Controls	Bruni	G

Rooms

Name	Building	Floor
DS1	OMI	Ground
N3	OMI	Ground
G	Pincherle	First

- “Retrieve all the lectures that are held on the ground floor"

SQL, an interactive language

```
SELECT Lecture, Room, Floor
FROM Rooms, Lectures
WHERE Name = Room
   AND Floor = 'Ground'
```

Lecture	Room	Floor
Systems	N3	Ground
networks	N3	Ground

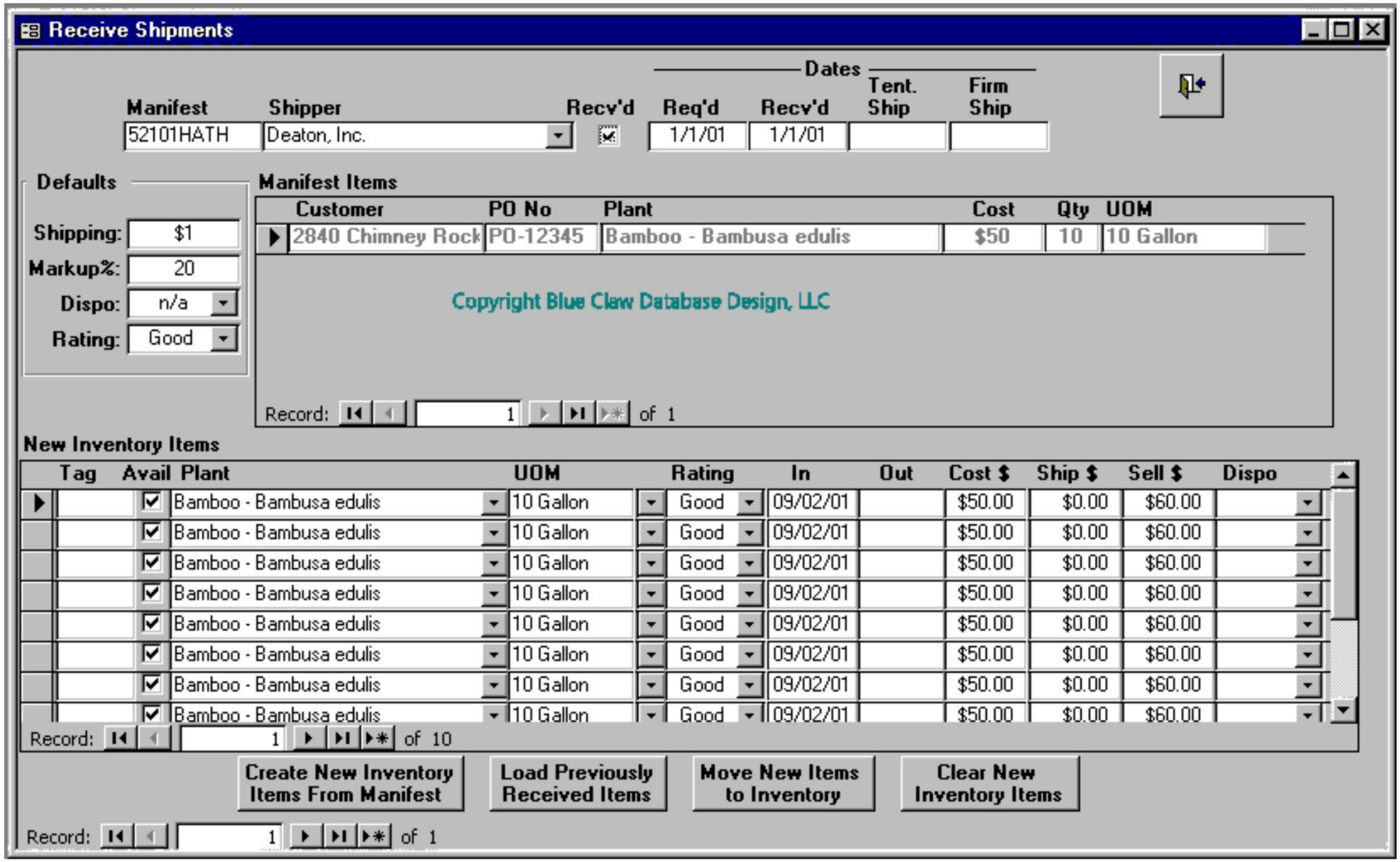
SQL injected in a host language

```
write('name of the city?'); readln(citta);
EXEC SQL DECLARE P CURSOR FOR
      SELECT NAME, INCOME
      FROM PEOPLE
      WHERE CITY = :city ;
EXEC SQL OPEN P ;
EXEC SQL FETCH P INTO :name, :income ;
while SQLCODE = 0 do begin
    write('name of the person:', name, 'rise?');
    readln(rise);
    EXEC SQL UPDATE PEOPLE
      SET INCOME = INCOME + :rise
                WHERE CURRENT OF P
    EXEC SQL FETCH P INTO :name, :income
  end;
EXEC SQL CLOSE CURSOR P
```

Ad Hoc SQL (Oracle PL/SQL)

```
declare  Stip number;
begin
     SELECT SALARY INTO STIP FROM EMPLOYEE
     WHERE NUMBER = '575488' FOR UPDATE OF SALARY;
     if Stip > 30 then
        UPDATE EMPLOYEE SET SALARY = SALARY * 1.1
      WHERE NUMBER = '575488';
     else
        UPDATE EMPLOYEE SET SALARY = SALARY * 1.15
      WHERE NUMBER = '575488';
    end if;
    commit;
exception
    when no_data_found then
      INSERT INTO ERRORS
      VALUES('NUMBER DOES NOT EXIST',SYSDATE);
end;
```

User friendly interface



A remark
(separating data from software)

- data manipulation language (DML)
  query and update (instances of) databases
- data definition language (DDL)
  is used for defining the schemas (logical, external, physical) and other operations

A DDL operation (over the schema)

```
CREATE TABLE hours (
  course CHAR(20) ,
  teacher      CHAR(20) ,
  room         CHAR(4)  ,
  hour         CHAR(5)  )
```

Actors and Users

- DBMS designers and developers
- database designers and database admins (DBA)
- software designer and developers of end-user applications
- users
  - final users (terminal operators): they execute pre-defined sets of operations (transactions)
  - end/casual user: they execute operations that are not defined a-priori, by using interactive languages

Database administrator (DBA)

- A specific person or a group of people that are in charge of the controlling and handling the database in a centralized way (e.g. efficiency, reliability, permissions)
- A *DBA* often also designs the database, except for some complex projects

Transactions (user's point of view)

- Any software implementing frequent and well known operations, defined a-priori, with very few expected exceptions.
- Examples:
  - paying money into one's account
  - issuing a birth certificate
  - a registration at the registry office
  - flight booking
- The transactions are implemented in a specific host language (either well known or "ad hoc")

Transactions, two point of views

- The user's:
  - any available software implementing an useful functionality/operation
- The system's:
  - unbreakable sequence of operations (see reliability)

DBMS: pros

Pros

- data as a shared resource, data bases modelling the environment
- standardizable and scaling up centralised data management
- provides integrated services
- reduces data redundancies and inconsistencies
- Independence of data (supporting the development and management of software applications)

DBMS: cons

Cons

- products can be costly as well as adopting such solutions
- functionalities are not detachable (the efficiency is reduced)